



Shaheed Zulfikar Ali Bhutto Institute of Science & Technology

COMPUTER SCIENCE DEPARTMENT

Total Marks: _____

Marks Obtained: _____

Applied Data Mining

Assignment # 03

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Question 1

Define Regression Analysis

Regression Analysis is a statistical and machine learning technique used to study the relationship between variables. It helps predict the value of one variable based on one or more other variables.

Regression is commonly used for forecasting, trend analysis, and decision-making.

Dependent Variable

A dependent variable is the variable whose value is predicted or explained by another variable.

- Also known as the **target variable** or **output variable**.
- Usually represented by **Y**.

Independent Variable

An independent variable is the variable used to predict or explain the dependent variable.

- Also known as the **predictor variable** or **input variable**.
- Usually represented by **X**.

Real-Life Example

Suppose a company wants to predict an employee's salary based on years of experience.

Years of Experience (X)	Salary (Y)
1	30,000
3	45,000
5	60,000

Here:

- **Independent Variable (X):** Years of Experience
- **Dependent Variable (Y):** Salary

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The salary depends on the years of experience.

Question 2

Types of Regression

Feature	Simple Linear Regression	Multiple Linear Regression	Logistic Regression
Number of Independent Variables	One	Two or More	One or More
Output Type	Continuous Value	Continuous Value	Categorical Value
Equation	$Y = a + bX$	$Y = a + b_1X_1 + b_2X_2 + \dots$	Uses Logistic Function
Purpose	Predict numerical values	Predict numerical values using multiple factors	Predict categories

1. Simple Linear Regression

Uses one independent variable to predict a dependent variable.

Example:

Predict salary using years of experience.

Equation:

$$[Y = a + bX]$$

2. Multiple Linear Regression

Uses two or more independent variables.

Example:

Predict house price using:

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- Area
- Number of bedrooms
- Location score

Equation:

$$Y = a + b_1X_1 + b_2X_2 + b_3X_3$$

3. Logistic Regression

Used when the output belongs to categories.

Example:

- Pass / Fail
- Yes / No
- Disease / No Disease

It predicts the probability of belonging to a particular class.

Question 3

Given Regression Model:

$$y = 108.35x + 60919.44$$

(a) When $x = 1$

Substituting $x = 1$:

$$y = 108.35(1) + 60919.44$$

$$y = 108.35 + 60919.44$$

$$y = 61027.79$$

Answer:

$$y = 61027.79$$

(b) When $x = 3$

Substituting $x = 3$:

$$y = 108.35(3) + 60919.44$$

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$$y = 325.05 + 60919.44$$
$$y = 61244.49$$

Answer:

$$y = 61244.49$$

Question 4

K-Nearest Neighbor (K-NN) Algorithm

K-Nearest Neighbor (K-NN) is a supervised machine learning algorithm used for classification and regression problems.

The algorithm classifies a new data point based on the classes of its nearest neighbors.

Working of K-NN

1. Choose the value of **k**.
2. Calculate the distance between the query point and all training points.
3. Select the **k nearest neighbors**.
4. Determine the majority class among those neighbors.
5. Assign that class to the query point.

How Similarity is Measured

Similarity is usually measured using distance metrics.

The most common distance metric is **Euclidean Distance**.

Formula:

$$Distance = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Smaller distance means greater similarity.

Role of Parameter k

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The parameter **k** represents the number of nearest neighbors considered during classification.

Small k (e.g., k = 1)

- Sensitive to noise
- May overfit

Large k (e.g., k = 10)

- More stable
- May underfit

A suitable value of k improves prediction accuracy.

Question 5

Given Dataset

Student	Math	CS	Result
1	4	3	Fail
2	6	7	Pass
3	7	8	Pass
4	5	5	Fail
5	8	8	Pass

Query Point

[X = (6,8)]

k = 3

Step 1: Calculate Euclidean Distances

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Step 1: Calculate Euclidean Distances

Student 1 (4,3)

$$d = \sqrt{(6 - 4)^2 + (8 - 3)^2}$$

$$d = \sqrt{4 + 25}$$

$$d = \sqrt{29}$$

$$d = 5.39$$

Student 2 (6,7)

$$d = \sqrt{(6 - 6)^2 + (8 - 7)^2}$$

$$d = \sqrt{0 + 1}$$

$$d = 1$$

Student 3 (7,8)

$$d = \sqrt{(6 - 7)^2 + (8 - 8)^2}$$

$$d = \sqrt{1 + 0}$$

$$d = 1$$

Student 4 (5,5)

$$d = \sqrt{(6 - 5)^2 + (8 - 5)^2}$$

$$d = \sqrt{1 + 9}$$

$$d = \sqrt{10}$$

$$d = 3.16$$

Student 5 (8,8)

$$d = \sqrt{(6 - 8)^2 + (8 - 8)^2}$$

$$d = \sqrt{4 + 0}$$

$$d = 2$$

Step 2: Distance Table

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Student	Distance	Result
2	1.00	Pass
3	1.00	Pass
5	2.00	Pass
4	3.16	Fail
1	5.39	Fail

Step 3: Select k = 3 Nearest Neighbors

Student	Distance	Result
2	1.00	Pass
3	1.00	Pass
5	2.00	Pass

Step 4: Majority Voting

Among the 3 nearest neighbors:

- Pass = 3
- Fail = 0

Majority Class = **Pass**

Final Prediction

Result = Pass



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Therefore, using the **K-NN algorithm with $k = 3$** , the query point (**Math = 6, CS = 8**) is classified as:

PASS